TIM-LOUIS WILKEN

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Profile: forecasting and optimisation of power markets; Python scripting, SQL, R.

EDUCATION

UNIVERSITY COLLEGE LONDON (UCL) / UNIVERSITY OF OXFORD — London / Oxford, United Kingdom
M.Sc. Energy Systems and Data Analytics (UCL); thesis as visiting student (Oxford) — *Sep 2025 – Sep 2026*

- **Average 76.8/100** across all taught modules (Distinction ≥ 70; dissertation pending). Top marks: Statistics for Energy Analytics **89**, Energy and Transport Analytics **85**, Econometrics of Energy Markets **82**.
- Thesis, written at Oxford (Jun–Sep 2026, fee waiver), supervised by A. Stratigakos (UCL), T. Morstyn and Y. Zhou (Oxford): *Deadline Commitments for GPU Jobs under a Forecast-Uncertain Non-Firm Grid Connection*. Two-stage stochastic MILP with ECMWF ensembles and chance constraints; 6–21 pp more deadlines than a greedy scheduler.

TECHNICAL UNIVERSITY OF MUNICH (TUM) — Munich, Germany
B.Sc. Management and Technology — *Oct 2021 – Mar 2025*

- Final grade 2.2 (German scale, 1.0 = best); electrical engineering, business analytics and finance; exchange semester at Singapore Management University (2023).
- Thesis (grade 1.3), *Rising connection requests for BESS in Germany*: analysed 2024 German day-ahead price volatility and battery cost degression behind 340 GW of BESS grid-connection requests, quantifying arbitrage revenues.

SIMON-MARIUS-GYMNASIUM — Gunzenhausen, Germany
Abitur; DPG-Abiturpreis for outstanding achievement in physics — *2012 – 2020*

TECHNICAL PROJECTS

- **GB Day-Ahead Price Forecasting**: online ensemble of 13 models (LSTM, Transformer, gradient boosting, LEAR and others), retrained daily to track changing market conditions and combined via Bernstein Online Aggregation; 21 % lower MAE than the LEAR benchmark over 2024–2026 (Diebold–Mariano $p < 0.001$); calibrated probabilistic forecasts; in a day-ahead battery backtest (50 MW / 100 MWh) it captures 93.7 % of the perfect-foresight arbitrage revenue in 2026 (93.1 % in 2025), vs. 90.4 % for LEAR and 76.5 % for persistence; strict leakage audit (pre-auction data only). *(Python, PyTorch, scikit-learn; live daily at* timlouiswilken.github.io/gb-dayahead-forecast*)*
- **BESS Trading Simulator**: web app for GB battery trading across day-ahead, intraday and imbalance markets on live Elexon BMRS data, with balancing-mechanism acceptances, a risk book and a P&L dashboard. *(React, TypeScript; built with a fellow student, 2026;* bess-trading-simulator.com*)*
- **Australian Electricity-Market Dynamics**: three-regime threshold vector error-correction (TVECM) analysis of how battery storage shapes SA–VIC wholesale price spillovers (52,608 half-hourly observations, 2SLS). *(Python, statsmodels;* github.com/TimLouisWilken/analysis-of-the-australian-electricity-market*)*

PROFESSIONAL EXPERIENCE

JOH. BERENBERG, GOSSLER & CO. KG — Munich, Germany
Origination and Growth Intern (full-time) — *May 2025 – Aug 2025*

- Built data-driven lead-generation workflows across 5 client segments, scaling to ∼100 qualified leads per week.
- Designed the segmentation and prioritisation logic behind the pipeline, improving lead quality and conversion.

LUX-WERFT UND SCHIFFFAHRT GMBH — Pleinfeld, Germany
Ship Operator, MS Brombachsee, Europe's largest inland passenger trimaran (part-time) — *Jun 2024 – Aug 2025*

MOONROC ADVISORY PARTNERS GMBH — Munich, Germany
Strategy Consulting Intern (full-time) — *Feb 2024 – May 2024*

- Quantified interview data for strategy workshops; co-authored *Retail Banking Compass 2024* (fintech chapter).

KPMG AG (AUDITING & ADVISORY) — Munich, Germany
Research Assistant, Audit (full-time) — *May 2023 – Aug 2023*

- Developed a client toolkit for sustainability reporting with TUM's Chair of Financial Accounting.

WEBSHAPE WEBDESIGN — Munich, Germany
Co-Founder & Managing Director (self-employed) — *Mar 2020 – Oct 2022*

- Co-founded a profitable web-design agency with high five-figure revenue, delivering 35+ client projects.

AWARDS & LEADERSHIP

- **Rolls-Royce University Project** (2025–26): reactor systems design (HTGR); team awarded first place.
- **Cambridge Climate Challenge** (2026): team lead of a three-person interdisciplinary team; finalist.
- **UCL Energy Society** (2025–26): External Speaker Officer, sourcing industry experts for panels and keynotes.

SKILLS

Energy markets	Day-ahead, intraday and imbalance markets (GB, DE); BESS arbitrage; Elexon and ECMWF data.
Methods	Statistical modelling, forecasting; stochastic & MILP optimisation; risk management; econometrics; ML.
Tools	Python (pandas, PyTorch, scikit-learn, statsmodels, PuLP); HiGHS, Gurobi; SQL, R, Git, TypeScript.
Languages	German (native), English (C1).